

Weinberg Angle and Electroweak RG Correction

David A. Besemer^{1,*}

¹*Independent Researcher*

(Dated: June 29, 2026)

We derive the weak mixing angle $\sin^2\theta_W$ from the tree-level cone geometry of SDGFT. At the Planck scale, the angle is exactly $1/9$, corresponding to the ratio of the cone half-angle to the total projection space. We calculate the 1-loop and 2-loop Renormalization Group (RG) corrections to the electroweak sector.

I. INTRODUCTION

II. THEORETICAL FORMULATION

Here we formulate the core mathematical relations governing the system:

$$\sin^2\theta_W^{(0)} = \left(\frac{\theta_{\max}}{90^\circ}\right)^2 = \frac{1}{9} \quad (1)$$

$$\sin^2\theta_W(M_Z) = \frac{1}{9} + \gamma_{EW} \approx 0.23122 \quad (2)$$

III. ELECTROWEAK RUNNING

The RG correction γ_{EW} represents the differential running of the electroweak parameters.

IV. CONCLUSION

The weak mixing angle is the primary parameter of the electroweak sector. We derive its high-energy boundary value and its running to the Z-boson mass scale.

The predicted value $\sin^2\theta_W \approx 0.23122$ is in perfect agreement with experimental data.

-
- [1] D. A. Besemer, “Axiomatic Foundations of SDGFT,” SDGFT-2026-01 (in preparation).
[2] D. A. Besemer, “Effective Spectral Dimension and the Banach Fixed-Point Theorem in SDGFT,” SDGFT-2026-04.

- [3] D. A. Besemer, `sdgft` Python package, <https://github.com/david-besemer/sdgft> (2026).

* david.besemer@sdgft.org